Guangyan Cai

Member of Technical Staff, World Labs
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EDUCATION

2020 – 2025 **University of California, Irvine** Irvine, CA
Ph.D. in Computer Science

- Advisor: Shuang Zhao
- Dissertation: *Physics-Based Inverse Rendering* (2025)
- M.S. in Computer Science conferred en route, 2025
- Dean's Award, Donald Bren School of Information and Computer Sciences, 2020

2016 – 2020 **University of California, San Diego** La Jolla, CA
B.S. in Computer Science

- Advisor: Ravi Ramamoorthi
- Cumulative GPA: 3.6 / 4.0 (Major GPA: 3.9 / 4.0)
- Undergraduate Award for Excellence in Research, 2020; Provost's Honors, 2016 – 2020

EXPERIENCE

Jul 2026 – Present **World Labs** New York, NY
Member of Technical Staff

- Working on simulation for robotic learning and embodied AI.

Aug 2025 – Jul 2026 **SceniX** New York, NY
Research Engineer

- Developed simulation engines for robotic learning systems, building photorealistic digital twins for robot training and evaluation.
- Co-authored a real-to-sim robot policy evaluation paper accepted to ICRA 2026.
- SceniX was acquired by World Labs in July 2026.

Jun – Sep 2023 **Adobe Research** San Jose, CA
Research Scientist Intern

- Proposed a lightweight, parallax-aware non-distant environment map for representing both distant and near-field illumination.
- Designed a pipeline that jointly reconstructs the shape and material of glossy objects and their surrounding lighting with physics-based differentiable rendering (PBIR-NIE).

Jun – Sep 2022 **Meta Reality Labs** Redmond, WA
Research Scientist Intern

- Investigated baking artifacts in material reconstruction with inverse rendering and proposed a method to mitigate them.
- Helped build Neural-PBIR, a hybrid pipeline combining neural reconstruction and physics-based differentiable rendering; co-first author of the ICCV 2023 paper.
- Showcased our reconstruction results at Meta Connect 2022.
- Neural-PBIR is released as a 3D reconstruction baseline in Meta's Digital Twin Catalog (CVPR 2025) and ranks first on the novel-scene relighting track of the Stanford-ORB benchmark.

PUBLICATIONS

2026 Real-to-Sim Robot Policy Evaluation with Gaussian Splatting Simulation of Soft-Body Interactions
Kaifeng Zhang, Shuo Sha, Hanxiao Jiang, Matthew Loper, Hyunjong Song, **Guangyan Cai**, Zhuo Xu, Xiaochen Hu, Changxi Zheng, and Yunzhu Li. *IEEE International Conference on Robotics and Automation (ICRA)*.

2025 **Image-Space Adaptive Sampling for Fast Inverse Rendering**
Kai Yan, Cheng Zhang, Sébastien Speierer, **Guangyan Cai**, Yufeng Zhu, Zhao Dong, and Shuang Zhao. *ACM SIGGRAPH Conference Papers*.

2025 **A Survey on Physics-based Differentiable Rendering**
Yunfan Zeng, **Guangyan Cai**, and Shuang Zhao. *arXiv:2504.01402*.

2024 **PBIR-NIE: Glossy Object Capture under Non-Distant Lighting**
Guangyan Cai, Fujun Luan, Miloš Hašan, Kai Zhang, Sai Bi, Zexiang Xu, Iliyan Georgiev, and Shuang Zhao. *arXiv:2408.06878*.

2023 **Neural-PBIR Reconstruction of Shape, Material, and Illumination**
Cheng Sun*, **Guangyan Cai***, Zhengqin Li, Kai Yan, Cheng Zhang, Carl Marshall, Jia-Bin Huang, Shuang Zhao, and Zhao Dong. *IEEE/CVF International Conference on Computer Vision (ICCV)*. *Equal contribution.

2022 **Physics-Based Inverse Rendering Using Combined Implicit and Explicit Geometries**
Guangyan Cai, Kai Yan, Zhao Dong, Ioannis Gkioulekas, and Shuang Zhao. *Computer Graphics Forum*.

2021 **Differentiable Time-Gated Rendering**
Lifan Wu*, **Guangyan Cai***, Ravi Ramamoorthi, and Shuang Zhao. *ACM Transactions on Graphics*. *Equal contribution.

2020 **Analytic Spherical Harmonic Gradients for Real-Time Rendering with Many Polygonal Area Lights**
Lifan Wu, **Guangyan Cai**, Shuang Zhao, and Ravi Ramamoorthi. *ACM Transactions on Graphics*.

PROJECTS

irtk uci-rendering.github.io/irtk
Inverse Rendering Toolkit: a Python library built on PyTorch for building inverse rendering pipelines, with a unified front end over multiple differentiable renderers. Used across several research projects.

isoext github.com/GuangyanCai/isoext
A Python/C++ library of GPU isosurface extraction algorithms such as Marching Cubes, using Thrust for GPU acceleration and nanobind for Python bindings.

psdr-jit github.com/andyyankai/psdr-jit
Contributor to psdr-jit, a physics-based differentiable renderer written in C++ with a Python interface.

REVIEWING

2026 ACM SIGGRAPH Technical Papers
2025 IEEE Transactions on Visualization and Computer Graphics
2024 The Visual Computer
2022 Eurographics

TEACHING

Fall 2021	CS112 Introduction to Computer Graphics	TA, UCI
Spring 2021	CS143A Principles of Operating Systems	TA, UCI
Winter 2021	CS143B Project in Operating System Organization	TA, UCI
Fall 2020	CS143B Project in Operating System Organization	Reader, UCI
Spring 2020	CSE168 Computer Graphics II: Rendering	Tutor, UCSD
2019 – 2020	CSE167 Computer Graphics	Tutor, UCSD